

ETAM Core Formulas × L0

Schrödinger Equation: Mathematical Linkage Notes

Positioning: This file is a **supplementary mathematical note** to the ETAM v2.0 document family. Goal: To carry out step-by-step mathematical linkage and interpretation between the **core formulas** of the ETAM mathematical framework and the **L0 Schrödinger equation**. Scope: Aligned only with **L0 (atomic scale)**; every instance of “1” is written with “characteristic function first, numbers second” .

0. Overview: Why Linkage Is Possible

ETAM § 2.1 treats A/T/E as “three-state projections of one and the same substance” ; the Schrödinger wavefunction ψ happens to encode all three states **simultaneously**:

$$\psi(\mathbf{r}, t) \left\{ \begin{array}{ll} \text{A (spatial state):} & \rho(\mathbf{r}) = |\psi(\mathbf{r}, t)|^2 \quad (\text{spatial distribution}) \\ \text{T (temporal state):} & e^{-iE_n t/\hbar} \quad (\text{time-dependent phase evolution}) \\ \text{E (energy state):} & E_n = \langle \psi_n | \hat{H} | \psi_n \rangle \quad (\text{eigenenergy}) \end{array} \right.$$

That is, ψ is a **single object that carries the three A/T/E projections at once**, which is isomorphic to ETAM’ s premise of “three-state projections of one substance, not three independent coordinates” . Four main threads of linkage:

1. **Normalization equality** $\int |\psi|^2 d\tau = 1 \leftrightarrow$ ETAM state normalization and reference ceilings $C_{*,\max}$
2. **Imaginary-axis engine** TDSE’ s $i \leftrightarrow$ the imaginary axis i of ETAM’ s complex port
3. **Fixed point** normalized ground state $\psi_0 \leftrightarrow$ ETAM synergetic fixed point $f = 1 + i0$

4. **Single-object three-projection** $\psi \leftrightarrow$ the mathematical semantics of the synergetic modulation operator “ $\cdot$ ”

Below we pin each core formula to Schrödinger.

1. ETAM Core Formula List (pending linkage)

No.	Formula	Meaning
F1	$Y = A \cdot T \cdot E$	Synergetic modulation output (“ $\cdot$ ” is not arithmetic multiplication)
F2	$f(R) = x(R) + i c(R)$	Left-hand complex-valued mapping
F3	$x = \langle C_{\text{raw}} \rangle, c = \langle \ M - 1 \ \rangle$	Dual-domain self-consistency
F4	$\tilde{a} = C_A / C_{A,\text{max}}, \tilde{t} = C_T / C_{T,\text{max}}, \tilde{o} = C_E / C_{E,\text{max}}$	Independent normalization of the three states
F5	$C_{\text{raw}} = \tilde{a}^{w_a} \tilde{t}^{w_t} \tilde{o}^{w_e}$	Fixed-structure-weight geometric synergetics
F6	$m_\ell = 1/\tilde{\ell}, c_\ell = m_\ell - 1 $	Modulation factor and modulation effort
F7	$f^{(0)} = C_{\text{raw}}^{(0)} + i c^{(0)} e^{i\theta_c^{(0)}}$	Equation One (L0 port)
F8	$d(\text{modulation})/d(\text{base}) = D e^{i\theta_c}$	θ_c phase differential
F9	$\Delta\omega = -\eta C_{\text{sys}}(1 - 2\tilde{a}) \omega$	Slow-variable feedback (A slot)

2. Thread 1: Normalization Equality ↔ Three-State Reference and F4

2.1 Born Normalization Gives “1”

$$\langle \psi | \psi \rangle = \int_{\mathbb{R}^3} |\psi(\mathbf{r})|^2 d\tau = 1$$

This is the physical origin of ETAM’ s “divide by each state’ s own ‘1’ ” : it unifies different states onto the unit scale of one and the same entity. **Explicit check (hydrogen 1s):**

$$\psi_{100}(r) = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0} \right)^{3/2} e^{-r/a_0}$$

$$\int |\psi_{100}|^2 d\tau := \frac{4}{a_0^3} \int_0^\infty r^2 e^{-2r/a_0} dr := \frac{4}{a_0^3} \cdot \frac{2}{(2/a_0)^3} := \frac{4}{a_0^3} \cdot \frac{a_0^3}{4} = 1$$

2.2 Three-State Reference Ceilings Computed Directly from ψ (denominator of F4)

- **A (spatial state):** Angular decomposition of electron density
 $\int |Y_{lm}|^2 d\Omega = 1$ split by $l = 0, 1, 2, 3$ into s, p, d, f . Slots $C_A = \sum \omega^{(p)} k(p)$, Pauli half-filled reference $s^1 p^3 d^5$:

$$C_{A,\max}^{(0)} = 1 \cdot 1 + 3 \cdot 2 + 5 \cdot 3 = 22$$

- **T (temporal state):** Atomic time unit from the time-dependent Schrödinger scale

$$\tau_a = \frac{\hbar}{E_h} = 2.418884 \times 10^{-17} \text{ s}, \quad C_{T,\max}^{(0)} = 2(1 + 2 + 3) = 12$$

- **E (energy state):** Eigenenergy ladder (Rydberg six-fold division)

$$\delta = \frac{E_h}{12} = 218.8 \text{ kJ} \cdot \text{mol}^{-1}, \quad C_{E,\max}^{(0)} = 2 \times 10 = 20$$

Thus, purely mathematically, F4’ s $\tilde{a} = C_A/22$, $\tilde{t} = C_T/12$, $\tilde{o} = C_E/20$ amounts to “divide the ψ -derived density/time/energy readings by the ‘1’ defined by Born normalization” .

3. Thread 2: TDSE’ s Imaginary Axis i ↔ ETAM Complex Port (linkage to F2, F7)

TDSE:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

Here i **carries the phase/dynamics engine**—the phase amount by which a state deviates from steady state is proportional to dynamical tension.

ETAM complex-valued mapping (F2) and Equation One (F7):

$$f(R) = x(R) + i c(R), \quad f^{(0)} = C_{\text{raw}}^{(0)} + i c^{(0)} e^{i\theta_c^{(0)}}$$

Linkage reasoning: Both place “the tension of deviating from ideal self-consistency” on the **imaginary axis**. - Write ψ as magnitude $\times$ phase: $\psi = |\psi| e^{i\phi}$; the real magnitude $|\psi| \leftrightarrow$ ETAM real-part coordination $x = C_{\text{raw}}$; - the phase ϕ and its deviation $\leftrightarrow$ ETAM imaginary part $c e^{i\theta_c}$. - A step further: the global time-dependent phase factor $e^{-iE_n t/\hbar}$ and ETAM’ s rotation factor $e^{i\theta_c}$ share the same form—both express “phase rotation around a fixed point” .

Hence TDSE’ s i and ETAM’ s imaginary axis i **may be regarded as sharing a common origin**: both use the complex imaginary part to express “the dynamical tension of deviating from the real equilibrium state” .

4. Thread 3: Fixed Point $\psi_0 \leftrightarrow f = 1 + i0$ (linkage to F3, F9)

The Schrödinger ground state satisfies:

$$\langle \psi_0 | \psi_0 \rangle = 1, \quad E_0 = \min_n E_n = \langle \psi_0 | \hat{H} | \psi_0 \rangle$$

ETAM synergetic fixed point (F3):

$$x^{(0)} = C_{\text{raw}}^{(0)} \rightarrow 1, \quad c^{(0)} \rightarrow 0 \quad \Leftrightarrow \quad f^{(0)} = 1 + i0$$

Linkage reasoning: On entry, Born normalization brings each state onto “1” (real part $x \rightarrow 1$); on exit, self-consistency converges the port to $1 + i0$ (imaginary part $c \rightarrow 0$). The two share the same “1” as a closed-loop anchor.

Furthermore, ETAM’ s slow-variable feedback (F9) is isomorphic to **Schrödinger imaginary-time evolution**:

$$\psi(\tau) = e^{-\hat{H}\tau} \psi(0) := \sum_n c_n e^{-E_n \tau} \psi_n \xrightarrow{\tau \rightarrow \infty} c_0 \psi_0$$

Imaginary-time propagation relaxes any initial state to the ground state (the energy-minimum fixed point). ETAM' s feedback $\Delta\omega = -\eta C_{\text{sys}}(1 - 2\tilde{a})\omega$ is precisely a gradient flow driving $\tilde{a}$ toward 1, with the same objective of the fixed point “modulation effort $C_{\text{sys}} \rightarrow 0$ ” . The two **may be seen as corresponding, mathematically, to the same class of gradient relaxation.**

5. Thread 4: Single-Object Three-Projection $\psi \leftrightarrow$ Synergetic Modulation Operator “ $\cdot$ ” (linkage to F1, F5)

In ETAM' s $Y = A \cdot T \cdot E$, “ $\cdot$ ” is the **synergetic modulation operator** (not arithmetic multiplication), expressing “change one, and all change together” . Its mathematical semantics can be explained via ψ :

- ψ is a **single object**; $\rho(\mathbf{r})$, the phase $e^{-iE_n t/\hbar}$, and the eigenenergy E_n are its three projections;
- altering the spatial distribution $\Rightarrow$ the phase and energy are affected together (“change one, all change”)—exactly the synergetic semantics of “ $\cdot$ ” .

F5' s geometric synergetics

$$C_{\text{raw}} = \tilde{a}^{w_a} \tilde{t}^{w_t} \tilde{o}^{w_e}, \quad w_a + w_t + w_e = 1$$

can be read mathematically as: **take the (equal-weight) geometric mean of the normalized readings of ψ ' s three projections**, yielding a scalar “coordination value” . The geometric mean (rather than an arithmetic sum) ensures that any term $\rightarrow 0$ pulls the whole toward 0—corresponding to ψ collapsing as a whole once any one projection collapses, consistent with “one substance, three states fused together” . The default equal weight 1/3 reflects that, within ψ , $|\psi|^2$, phase, and energy are descriptions of **equal standing**, with no a-priori bias.

6. θ_c Phase Differential and Schrödinger Parameter Response (linkage to F8)

F8:

$$\frac{d(\text{modulation})}{d(\text{base})} = |D| e^{i\theta_c}, \quad |D| = 1$$

$|D| = 1$ gives a hard anchor on the real part (magnitude does not drift), while θ_c enters the imaginary part (phase rotates freely).

Linkage reasoning: In the Schrödinger system, Born normalization fixes the magnitude $\|\psi\| = 1$ (hard real-part anchor), whereas the state's response to parameters resides in the phase. The Hellmann–Feynman theorem

$$\frac{dE_n}{d\lambda} = \left\langle \psi_n \left| \frac{\partial \hat{H}}{\partial \lambda} \right| \psi_n \right\rangle$$

states that the energy (real part) has a definite derivative as parameters vary; it is the **phase response** carried by the wavefunction's differential $d\psi/d\lambda$ that constitutes the dynamical content—precisely the structure where θ_c enters the imaginary part: the real part is locked by normalization ($|D| = 1$), and the dynamics live entirely in the imaginary-phase part. The two structures are isomorphic.

7. Core Formula ↔ Schrödinger Linkage Table

ETAM core formula	Schrödinger counterpart	Linkage property
F4 normalization $\tilde{\ell} = C/C_{\max}$	$\int \psi ^2 d\tau = 1$	denominator is Born “1”
F1 “•” operator	ψ single-object three-projection	synergetic semantics share a common origin
F5 geometric synergetics C_{raw}	geometric mean of $ \psi ^2$ / phase / E_n	equal-weight, geometric mean
F2/F7 imaginary axis i	TDSE's i	imaginary part = dynamical tension
F3/F9 fixed point $f = 1 + i0$	normalized ground state ψ_0 / imaginary-time evolution	gradient relaxation to minimum
F8 θ_c differential	HF theorem / phase response	real part hard-anchored, phase into imaginary part

8. Mathematical Consistency Conclusions (L0 only)

1. ETAM' s “1” , purely mathematically at L0, = ψ (the characteristic function); Born normalization explicitly yields this “1” .
 2. The three-state reference ceilings 22/12/20 can be computed directly from ψ ' s angular decomposition, the atomic time unit, and the Rydberg ladder.
 3. The imaginary axis i of TDSE and the imaginary axis i of ETAM' s complex port are isomorphic (imaginary part = dynamical tension).
 4. The synergetic fixed point $f = 1 + i0$ is isomorphic to the normalized ground state plus imaginary-time relaxation (gradient flow to the minimum).
 5. The “ $\cdot$ ” operator and “ ψ as a single object carrying the A/T/E three projections” are consistent in mathematical semantics.
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9. Limitations and Open Items (discussion)

- The above “isomorphic / sharing a common origin” relations are mathematical-structure analogies, not strict derivational equivalences; ψ is the exact solution of a physical equation, whereas ETAM slots are a discretized encoding approximation—their scale and accuracy boundaries remain to be tested against more examples.
- The analogy between F5' s equal weight 1/3 and ψ ' s three descriptions being “of equal standing” belongs to an interpretive framework; when domain-specific structural weights override it, this analogy needs to be re-evaluated.
- The correspondence between imaginary-time evolution and the feedback loop is best verified numerically in follow-up work against concrete atoms (Na⁺/Cl⁻/O/H).

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